



**Year 11 Mathematics Extension 1
Preliminary Assessment Task 2
Term 3, Week 4, 2020**

Name: _____

Wednesday, 12th August 2020

- Start each question on a new page.
- NESA approved calculators are permitted in this task.
- Marks may be deducted for insufficient or illegible work.
- Total possible mark: **32**
- Time allowed: **45 minutes + 2 minutes reading time**
- For any multiple choice question, **write the letter** that corresponds to your answer on your writing paper.

Question 1 (8 marks)

Marks

- (a) What is the value of $f'(2)$ if $f(x) = \frac{1}{3x}$? **1**
- (A) $-\frac{1}{36}$
- (B) $-\frac{1}{12}$
- (C) $\frac{1}{6}$
- (D) $\frac{1}{3}$
- (b) Find the value of c such that $y = -10x + c$ is a tangent to the parabola $y = 3x^2 + 2x - 4$. **3**
- (c) Given that $y = (3x + 1)^2$. Show that $y - y' + 4 = x(y' - 12) - y$. **2**
- (d) (i) On a Cartesian plane, draw a neat sketch of the function of the function $y = f(x)$ **1**
where:
- $$f(x) = \begin{cases} x^2 - 4, & \text{for } x < 2 \\ 4x - 8, & \text{for } x \geq 2 \end{cases}$$
- (ii) Is the function above differentiable at $x = 2$? Justify your answer. **1**

End of Question 1

Question 2 starts on the next page

Question 2 (8 marks) *Start a new page*

Marks

- (a) For $x > 0$, which expression is NOT equivalent to $e^{\ln x}$? **1**
- (A) $\ln(e^x)$
- (B) $x^{\ln e}$
- (C) $(\ln e)^x$
- (D) $\frac{1}{e^{\frac{\ln 1}{x}}}$
- (b) Solve for x : **2**
- $$\log_3 x + 2 = \log_3(x + 2)$$
- (c) By using a suitable substitution, solve $4^x - 5 \times 2^{x+1} + 16 = 0$. **2**
- (d) Find the equation of the normal to the curve $y = e^{2x+2}$ at the point where $x = -1$. **3**

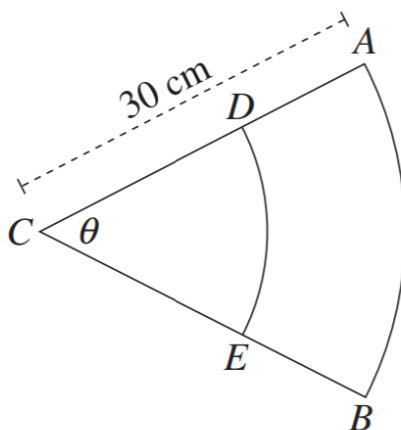
End of Question 2

Question 3 starts on the next page

Question 3 (7 marks) *Start a new page*

Marks

- (a) How many solutions does the equation $\sin^2 x + 3\sin x \cos x + 2\cos^2 x = 0$ have in the domain $0 \leq x \leq 2\pi$? **1**
- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (b) The region ABC is a sector of a circle with radius 30 cm, centred at C . The angle of the sector is θ . The arc DE lies on a circle also centred at C , as shown in the diagram. **2**



The arc DE divides the sector ABC into two regions of equal area.

Find the exact length of the interval CD .

Question 3 continues on the next page.

Question 3 (continued)

- (c) Two hundred rabbits in a region with an estimated population of 200 000 rabbits have a highly contagious disease.

The disease is known to spread at the weekly rate of 1% of the remaining healthy rabbits such that:

$$\frac{dP}{dt} = 0.01(200\,000 - P)$$

where P is the number of infected rabbits after t weeks.

- (i) Show that $P = 200\,000 - 199\,800e^{-0.01t}$ satisfies the differential equation. **1**
- (ii) How many days does it take for half of the rabbit population to become infected? **3**

End of Question 3

Question 4 starts on the next page

Question 4 (9 marks) *Start a new page***Marks**

- (a) The radius of a circle is decreasing at a constant rate of 0.1 cm/s. **1**

In terms of the circumference, C , what is the rate of change of the area of the circle in cm^2/s ?

(A) $-0.1C$

(B) $\frac{-0.1C}{2\pi}$

(C) $0.1^2 C$

(D) $0.1^2 \pi C$

- (b) Solve for $-\pi \leq x \leq \pi$: **1**

$$\tan\left(x - \frac{\pi}{6}\right) = \sqrt{3}$$

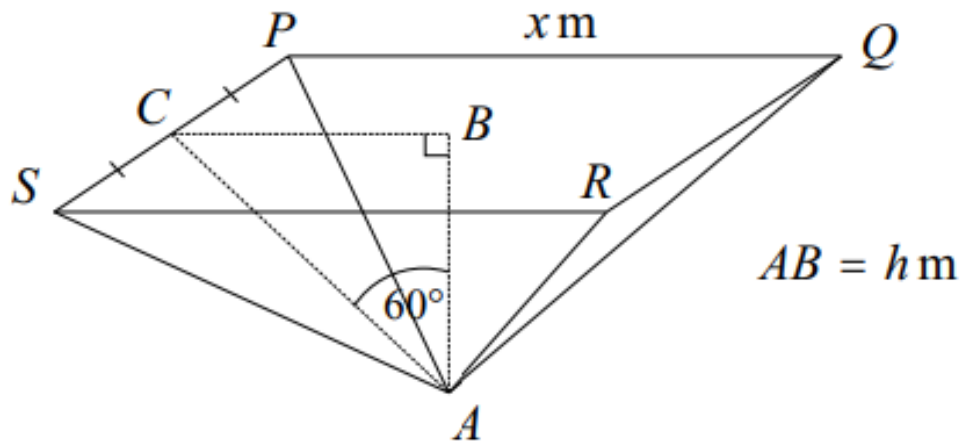
- (c) Solve the equation for $0 \leq \theta \leq 2\pi$, leaving your answer in terms of π . **3**

$$4\sin^2 \theta + 2\cos \theta = 3$$

Question 4 continues on the next page.

Question 4 (continued)

(d)



The diagram above shows a container in the shape of a square pyramid with ‘base’ $PQRS$.

The centre of the base is the point B , and the perpendicular height AB of the pyramid is h m. The side length of the square base is x m, and the slant height AC of the pyramid makes an angle of 60° to the vertical, as shown in the diagram.

- (i) Show that $x = 2h\sqrt{3}$. 1
- (ii) Hence show that $h = \left(\frac{V}{4}\right)^{\frac{1}{3}}$, where V is the volume of the pyramid in cubic metres. (Volume of pyramid: $V = \frac{1}{3} \times \text{area of base} \times \text{perpendicular height}$) 1
- (iii) The pyramid is initially empty and is filled with water at a constant rate of $0.5 \text{ m}^3/\text{min}$. Find that rate at which the height of the water is increasing after it has been filled for 8 minutes. 2

End of Question 4

End of Assessment



2020 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 1</u> a) $f(x) = \frac{1}{3}(x)^{-1}$ $f'(x) = -\frac{1}{3} \times \frac{1}{x^2}$ $f'(2) = -\frac{1}{3} \times \frac{1}{2^2} = -\frac{1}{12}$ Answer: B b) $y = 3x^2 + 2x - 4$ $y' = 6x + 4$ from $y = -10x + c$, $m = -10$ $y' = -10$ $-10 = 6x + 4$ $-12 = 6x$ $x = -2$ Sub $x = -2$ into y $y = 3(-2)^2 + 2(-2) - 4$ $y = 4$ Sub x & y into tangent $4 = -10(-2) + c$ $4 = 20 + c$ $c = -16$ c) $y = (3x+1)^2$ $y' = 2(3x+1) \times 3$ $= 6(3x+1)$ LHS / $(3x+1)^2 - 6(3x+1) + 4$ $9x^2 + 6x + 1 - 18x - 6 + 4$ $9x^2 - 12x - 1$	① for x ① for y ① correct sub of x & y ① correct working for LHS	RHS $x(18x+6-12) - 3(x+1)^2$ $18x^2 - 6x - 9x^2 - 6x - 1$ $9x^2 - 12x - 1$ $\therefore \text{LHS} = \text{RHS}$ d) i) ii) it is smooth @ $x = 2$ hence differentiable note: error carry for part (i) incorrect graph	① correct working for RHS. ① must be smooth @ $x = 2$ ① linear function must be smooth ① Having smooth



2020 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

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<u>Question 2</u> a) Answer: ($(\ln e^x) = 1/x$ b) $2 = \log_3(x+2) - \log_3 x$ $\log_3\left(\frac{x+2}{x}\right) = 2 \quad \checkmark$ $\frac{x+2}{x} = 9$ $x+2 = 9x$ $2 = 8x$ $x = \frac{1}{4} \quad \checkmark$ c) $4^x - 5 \times 2^{x+1} = 16$ let $u = 2^x$ $(2^x)^2 - 5 \times (2^x) \times 2 + 16 = 0$ $(2^x)^2 - 10(2^x) + 16 = 0 \quad \checkmark$ $u^2 - 10u + 16 = 0$ $(u-8)(u-2) = 0$ $u = 8 \quad \text{or} \quad u = 2$ $2^x = 8 \quad \quad \quad 2^x = 2$ $x = 3 \quad \quad \quad x = 1 \quad \checkmark$	One of by laws Answer Correct sub Answer note: no marks awarded for guess & check.	d) $y = e^{2x+2}$ $y' = 2e^{2x+2}$ at $x = -1$, $y' = 2e^{-2+2}$ $y' = 2e^0$ $y' = 2$ $m_1 = 2$, normal: $m_2 = -\frac{1}{2} \quad \checkmark$ @ $x = -1$, $y = e^0$ $y = 1$ $(-1, 1)$ with normal $= -\frac{1}{2} \quad \checkmark$ Equation of normal $y - 1 = -\frac{1}{2}(x + 1)$ $2y - 2 = -x - 1$ $x + 2y - 1 = 0 \quad \checkmark$	Finding normal On x, y coordinates Equn of normal.



2020 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 3</u> <u>a) Answer: D</u> $(\sin x + 2\cos x)(\sin x + \cos x) = 0$ $\sin x = -2\cos x \quad \sin x = -\cos x$ $\tan x = -2 \quad \tan x = -1$ 2 solutions 2 solutions b) Area of sector ACB $\frac{1}{2}r^2\theta$ $\frac{1}{2} \times (30)^2 \times \theta$ $= 450\theta$ Area of sector DCE $\frac{1}{2} \times 450\theta$ $= 225\theta$ $\therefore \frac{1}{2}r^2\theta = 225\theta$ $r^2 = 450$ $r = 15\sqrt{2}, r > 0$		c) i) $P = 200\,000 - 199\,800 e^{-0.01t}$ $\frac{dP}{dt} = -1998 e^{-0.01t}$ Given $\frac{dP}{dt} = 0.01(200\,000 - P)$ sub in P $0.01(200\,000 - (200\,000 - 199\,800 e^{-0.01t}))$ $0.01(199\,800 e^{-0.01t})$ $1998 e^{-0.01t} \quad \text{QED}$ ii) when $P > 100\,000$ $200\,000 - 199\,800 e^{-0.01t} > 100\,000$ $199\,800 e^{-0.01t} < 100\,000$ $e^{-0.01t} < \frac{100\,000}{199\,800} \quad \checkmark$ $-0.01t < \ln\left(\frac{500}{999}\right)$ $t > 100 \ln\left(\frac{500}{999}\right) \quad \checkmark$ $t > 69.214 \dots \text{ weeks}$ $t > 484.502 \dots \text{ days} \quad \checkmark$ $\therefore$ Half rabbit population is infected after <u>485</u> days. note: if answer written as 69.24 days but rounded to 70 days $\rightarrow$ mark still awarded.	Working * allow for alternate method. in terms of e for t answer no mark awarded for 484 days



2020 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 4</u> a) $\frac{dr}{dt} = -0.1 \text{ cm/s}$ $A = \pi r^2$ $\frac{dA}{dr} = 2\pi r$ $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ $= 2\pi r \times -0.1$ $= -0.1\pi$ Answer: A b) $\tan\left(\pi - \frac{\pi}{6}\right) = \sqrt{3}$ $\pi - \frac{\pi}{6} = \frac{\pi}{3} \quad \text{or} \quad -\frac{2\pi}{3}$ $\pi = \frac{\pi}{3} \quad \text{or} \quad -\frac{\pi}{3}$ ✓ c) $4\sin^2\theta + 2\cos\theta = 3$ $4(1 - \cos^2\theta) + 2\cos\theta = 3$ $4 - 4\cos^2\theta + 2\cos\theta = 3$ $4\cos^2\theta - 2\cos\theta - 1 = 0$ ✓ $\cos\theta = \frac{2 \pm \sqrt{4+16}}{8}$ $\cos\theta = \frac{2 \pm 2\sqrt{5}}{8} \Rightarrow \frac{1 \pm \sqrt{5}}{4}$ ✓ $\theta = \frac{\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}, \frac{9\pi}{5}$ ✓		d) $\tan\theta = \frac{\frac{1}{2}\pi}{h} \quad \frac{(BC)}{(BA)}$ $\sqrt{3} = \frac{\frac{1}{2}\pi}{h}$ $\sqrt{3}h = \frac{1}{2}\pi$ $\pi = 2\sqrt{3}h$ ✓ ii) $V = \frac{1}{3}\pi r^2 h$ $V = \frac{1}{3}(2\sqrt{3}h)^2 \times h$ $V = \frac{12h^2}{3} \times h$ $V = 4h^3 \Rightarrow \frac{V}{4} = h^3 \Rightarrow h = \left(\frac{V}{4}\right)^{\frac{1}{3}}$ iii) $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $\frac{dV}{dt} = 0.5$ $\frac{dh}{dV} = \frac{1}{3} \left(\frac{1}{4}V\right)^{-\frac{2}{3}} \times \frac{1}{4}$ $= \frac{1}{12 \left(\frac{1}{4}V\right)^{\frac{2}{3}}}$ ✓ after 8 mins: $V = 0.5 \times 8$ $V = 4$ $\frac{dh}{dt} = \frac{1}{12 \left(\frac{1}{4} \times 4\right)^{\frac{2}{3}}} \times 0.5$ $= \frac{1}{24} \text{ m/min}$ ✓	① working ① correct working to h. ① correct differentiation for $\frac{dh}{dV}$ ① answer

① answer

① in terms of $\cos^2\theta$

① using quadratic formula

① solution



2020 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Alternate Solution</u> $\frac{dv}{dt} = \frac{dv}{dh} \times \frac{dh}{dt}$ $0.5 = 12h^2 \times \frac{dh}{dt}$ Also when $t=8$, $v=4$ as $\frac{dv}{dt} = 0.5 \text{ m}^3/\text{min}$ h is $\left(\frac{4}{3}\right)^{1/3} = 1 \text{ m}$ so $\frac{dh}{dt} = \frac{1}{24h^2} = \frac{1}{24} \text{ m/min}$			